ANALOG CIRCUITS

Class Test 01 | Complete lecture-first study guide

What this guide does: Organizes all topics visibly taught in your 22 handwritten lecture pages; maps each topic to your supplied Chapter 2 textbook; explains the concepts, solved maths, waveform logic, practice questions, and a day-by-day plan.

> Exam note: Your handwritten class note is the controlling syllabus. Textbook material is included only where it explains or practises a topic present in the note. Some book topics are deliberately excluded.

The 7 high-yield skill groups

01 Diode fundamentals, ON/OFF decision and load lines

02 Series, parallel and multiple-diode circuit calculations

03 Diode AND/OR logic and gate implementation

04 Half-wave, bridge and center-tapped rectifiers / transformer

05 Positive, negative, shunt, series and biased clippers

06 Clampers, capacitor charge and waveform shifts

07 Zener regulators and all three mathematical cases

Prepared from: “2026-11-11 15-51-36.pdf” (22 handwritten pages) and “CT01_C2.pdf” (74 pages, textbook Chapter 2: Diode Applications).

Source convention: “note p.” means PDF page of the 22-page notebook; “book p.” means printed textbook page (PDF page = printed page − 54).

1 | Exactly what to study, and what to skip

Handwritten note	Book: section / printed pages	Priority
pp. 1–3: load line; series diode; bias and KVL	§2.2 pp. 56–60; §2.3 pp. 61–66	Critical maths
pp. 4–5: parallel, LEDs, complex branches	§2.4 pp. 67–70	Critical maths
pp. 6–7: AND, OR, recognize gate	§2.5 pp. 70–72	Gate drawing + truth tables
pp. 8–14: clipper variants and outputs	§2.8 pp. 78–84	Waveform-heavy
pp. 15–16: clamper variants, voltage shifts	§2.9 pp. 85–88	Waveform-heavy
p. 17: half-wave rectifier	§2.6 pp. 72–75	Diagram + numerical
pp. 18–19: bridge and CT full-wave	§2.7 pp. 75–78	Diagram + PIV
pp. 20–22: Zener; 3 regulator cases	§2.11 pp. 91–98	Highest maths priority

Not on the teacher-note list: do not spend study hours here

§2.10 Networks with a DC and AC Source (pp. 88–90); §2.12 Voltage-Multiplier Circuits (pp. 98–100); §2.13 Practical Applications (pp. 101–110); §2.14 Summary (read only if useful); §2.15 Computer Analysis; full-page simulation exercises. Basic transformer voltage ratio is a short supporting idea, not a separate lecture unit in these notes.

> Exam note: The end-of-chapter problem heading calls Zener “2.10,” even though the main theory is §2.11. Use the printed problem numbers rather than relying on that inconsistent label.

The fastest efficient reading procedure

• Read the teacher’s page. Redraw every circuit and waveform without looking.

• Find the matching book example below; copy the method, not the wording.

• Solve the same example using different values before looking at an answer.

• Do one selected book exercise + two new practice questions.

• Keep a “mistake log”: wrong diode direction, missed voltage drop, wrong polarity, wrong peak/amplitude.

2 | Always start with these five tools

Rule / symbol	What it means
Ideal diode	ON → short circuit, $V_D = 0$ V. OFF → open circuit, $I_D = 0$.
Silicon (Si) diode	Approximate ON drop $V_D \approx 0.7$ V; reverse biased / below turn-on → OFF.
LED	Forward drop is color/type dependent. Use the value specified in the circuit; your notes use ≈2 V for colored LEDs.
Ohm's law	$I = V / R$; 1 mA through 1 kΩ gives a 1 V drop.
Kirchhoff (KVL / KCL)	Around a loop: sum of rises and drops = 0. At a junction: current entering = current leaving.
Open-circuit vs short	Open → zero current, possible nonzero voltage. Short → zero voltage, current depends on remainder.

Universal 6-step diode-analysis algorithm

- Mark supply polarities, diode anode/cathode, and defined output polarity.
- Assume which diodes are ON and OFF.
- Replace ON silicon diodes by 0.7 V drops (ideal: 0 V); replace OFF by an open circuit.
- Apply KVL, KCL and Ohm's law to the remaining circuit.
- Check your ON/OFF assumptions against the calculated voltages/currents.
- Report requested voltages, currents, units and their signs.

3 | Load-line analysis (teacher note p. 1)

The load line represents the limitations imposed by a battery E and resistor R. The diode characteristic is the nonlinear I–V curve. Their intersection is the operating, or Q-point, giving both V_{DQ} and I_{DQ}.



KVL: $E = V_D + I_D R \quad \Rightarrow \quad I_D = (E - V_D)/R$

Find the intercept	Set	Plot this coordinate
Vertical (current) axis	$V_D = 0$ V	$I_D = E/R$
Horizontal (voltage) axis	$I_D = 0$ A	$V_D = E$

How to draw: (1) Calculate E/R and E. (2) Place (V_D=0, I_D=E/R) and (V_D=E, I_D=0) on the diode-curve graph. (3) Join points with a straight line. (4) The crossing with the diode curve is the Q-point.

Worked example — E = 10 V, R = 0.5 kΩ

Vertical intercept I_D = 10/500 = 20 mA. Horizontal intercept V_D = 10 V. With a silicon 0.7-V model, I_D ≈ (10−0.7)/500 = 18.6 mA, V_R ≈ 9.3 V. With a drawn real diode curve, Q is read from the curve, not merely calculated using 0.7 V.

Matching book work: Example 2.1 (full curve), 2.2 (approximate), 2.3 (ideal); study book Figures 2.1–2.7 (printed pp. 56–60).

> Exam note: The note writes I_D maximum = E/R at V_D = 0 and V_D maximum = E at I_D = 0. These are LOAD-LINE intercepts—not necessarily the actual operating current or diode voltage.

4 | Series diode circuits (teacher pp. 2–3)

> **Si diode ON: I = (E − 0.7 V)/R, V_R = E − 0.7 V**

If an OFF diode is the only series path, the circuit current is zero; thus V_R = 0. The remaining supply voltage appears across the open diode with sign depending on the specified polarity.

Worked example A — directly from the class note, p. 2

E = 8 V; R = 2.2 kΩ; silicon D forward. Then V_D = 0.7 V; V_R = 8 − 0.7 = 7.3 V; I = 7.3/2.2k = 3.32 mA. Reverse the diode: I = 0; V_R = 0; magnitude |V_D| = 8 V.

Worked example B — two supplies, book Example 2.9; note p. 3

E_1 = 10 V, E_2 = −5 V, R_1 = 4.7 kΩ, R_2 = 2.2 kΩ; conducting Si diode: total driving magnitude is 10 + 5 − 0.7 = 14.3 V. Thus I = 14.3/6.9k = 2.07 mA, V_1 = I × 4.7k = 9.73 V; V_2 = I × 2.2k = 4.55 V. For the book's marked reference polarity, V_o = V_2 − 5 = −0.45 V.

Matching book work: Examples 2.4–2.9 (pp. 63–67), especially Examples 2.4, 2.5, 2.8, 2.9.

5 | Parallel and series-parallel diodes (teacher pp. 4–5)

Diodes in parallel share the same voltage. If two identical diodes conduct and the problem assumes equal current sharing, each carries half of the total. In a real circuit, sharing is not automatically equal unless device matching / balancing is justified; textbook examples explicitly approximate identical devices.

Worked example A — note p. 4; book Example 2.10

10 V source → 0.33 kΩ series resistor → two parallel conducting Si diodes. Both have the same ≈0.7-V forward drop. I_total = (10 − 0.7)/330 = 28.18 mA. For identical diodes, I_D1 = I_D2 ≈ 14.09 mA.

Worked example B — LED resistor in note p. 4

A green LED with a specified 2-V drop operates from an 8-V supply at 20 mA. Required resistor R = (8−2)/(0.020) = 300 Ω or 0.30 kΩ. The red and green LED branches must be checked for their directions and forward voltage assumptions.

Worked example C — mixed branches, class note p. 5

The note applies KCL: I_D2 = I_2 − I_1, with I_2 ≈ 3.32 mA and I_1 ≈ 0.212 mA, so I_D2 ≈ 3.11 mA. Practise drawing the branch current directions before subtraction.

Matching book: Examples 2.10–2.13 (pp. 67–70), particularly Fig. 2.28 and Fig. 2.37.

6 | Diode gates, AND and OR (teacher pp. 6–7)



Positive logic convention in your notes: logic 0 = 0 V, logic 1 = +5 V. “Logic output” is the interpreted high/low state; the analog diode output can be ≈0.7 V or ≈4.3 V rather than exactly 0 V or 5 V.

A	B	AND = A·B	OR = A+B
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

Implementing a diode AND gate

Diode cathodes point toward input nodes, anodes meet at output; a resistor pulls output UP to +5 V. Any LOW input forward-biases a diode and pulls output LOW (typically ≈0.7 V using Si). Only when A=B=HIGH can the output remain HIGH. Your note p. 7 asks you to identify exactly this orientation, with a pull-up supply.

Implementing a diode OR gate

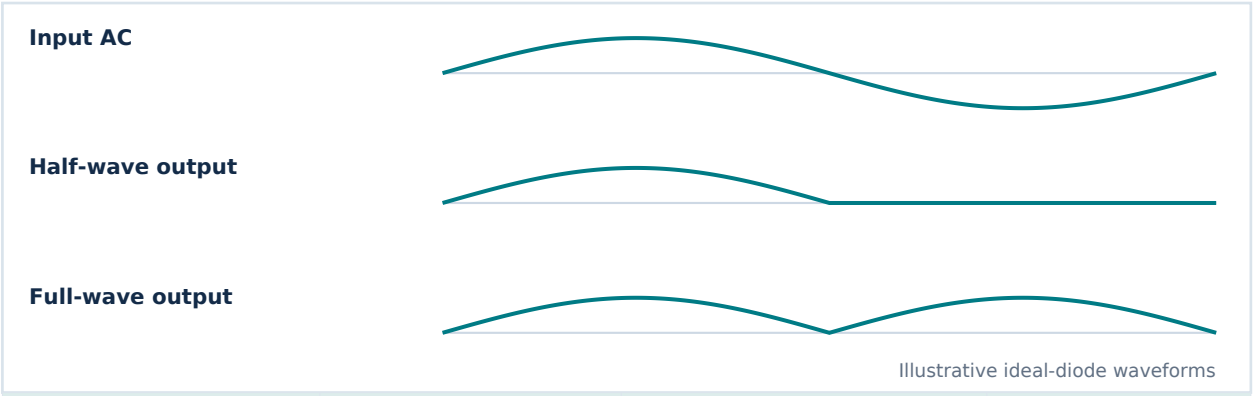
Diode anodes point toward input nodes, cathodes join at output; a resistor pulls output DOWN. Any HIGH input forward-biases its diode and raises output to about 5−0.7=4.3 V. Only A=B=LOW produces LOW output.

Matching book: §2.5, Examples 2.14 (OR) and 2.15 (AND), printed pp. 70–72. Book examples use +10 V logic levels; teacher uses +5 V. Same operation, different voltages.

> Exam note: Do not classify a gate solely by whether diodes “point left or right.” Check how the output node is biased (pull-up versus pull-down) and what happens when ONE input is LOW or HIGH.

7 | Rectifiers and the center-tapped transformer (teacher pp. 17–19)

A rectifier uses diodes to convert an alternating input into a unidirectional, pulsating output. The key examination tasks are identifying conducting diodes, sketching v_o, calculating average (DC) output and PIV.



Illustrative ideal-diode waveforms

Circuit	Ideal V_DC	Conduction	Diode PIV
Half-wave (one diode)	V_m/π ≈ 0.318 V_m	One half-cycle only	≥ V_m
Bridge full-wave (4 diodes)	2V_m/π ≈ 0.636 V_m	Two diodes per half-cycle	≥ V_m
CT full-wave (2 diodes)	2V_m/π ≈ 0.636 V_m	One diode per half-cycle	≥ 2V_m

Here V_m means the peak voltage across the relevant rectifier input: for the CT circuit it is the peak across each half of the center-tapped secondary. For the bridge it is the peak across the entire secondary. Misdefining V_m is a common exam mistake.

Half-wave rectifier — note p. 17

Positive input half: diode ON, output follows the positive half-wave. Negative half: diode OFF, output = 0. For ideal diodes, V_DC = V_m/π. Ripple pulses repeat at the input frequency f.

Bridge full-wave rectifier — note p. 18

Four diodes form a bridge: one diagonal pair conducts on the positive half; the opposite diagonal pair on the negative half. The load polarity stays the same. For a silicon bridge, two ON diodes cause an approximate 1.4-V total forward drop when conducting. Ripple frequency becomes 2f.

Center-tapped full-wave rectifier — note p. 19

Transformer secondary has two equal halves around a center tap. Diode D_1 conducts on one half-cycle, D_2 on the other; both feed the load with the same polarity. Only one silicon diode drop is in the active load path. However each diode must withstand about 2V_m PIV (where V_m is one half-secondary peak).

For an ideal transformer: $V_{s,rms}/V_{p,rms} = N_s/N_p$; $V_{peak} = \sqrt{2} \times V_{rms}$

Exam note: Your note explicitly calls the CT rectifier “better than bridge” in its PIV discussion. This should be read carefully: the note gives CT PIV = 2V_m, which is MORE reverse-voltage stress than bridge PIV = V_m for equal per-half peak definitions; the bridge also uses two forward diodes while CT uses one. Neither is universally “better.” Use the specific pros/cons and PIV formulas.

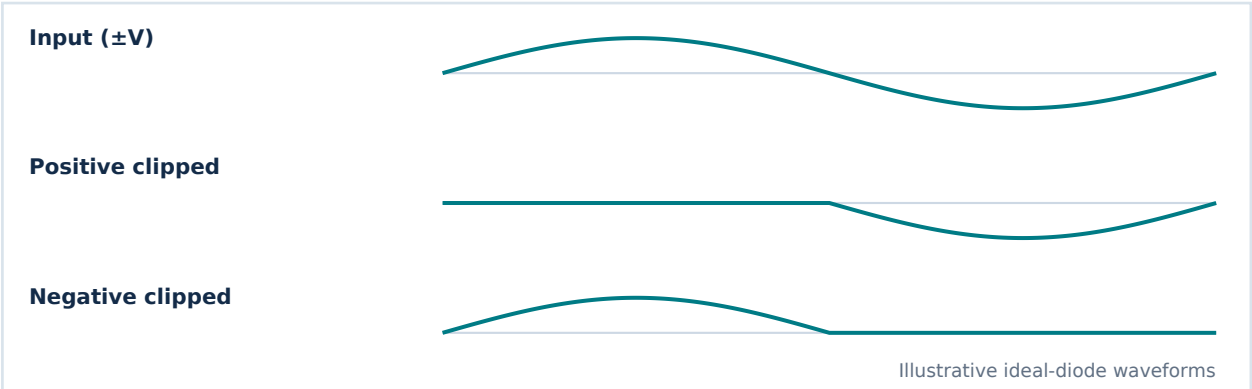
Worked numerical comparison

If a 12-V rms AC applied directly to a rectifier input has peak $V_m = 12\sqrt{2} = 16.97$ V, an ideal half-wave rectifier averages 5.40 V; an ideal bridge full-wave averages 10.80 V. Bridge diode PIV must be at least 16.97 V (normally choose a safety margin). If 12 V rms were the full CT secondary rating, each half would have 6 V rms, so its V_m and output would be different.

Matching book: §2.6 pp. 72–75; §2.7 pp. 75–78. Trace Figures 2.44–2.63; pay extra attention to Figures 2.57, 2.60 and 2.63.

8 | Clippers — series, shunt, positive, negative, biased (teacher pp. 8–14)

A clipper deliberately removes, blocks or limits part of the input waveform. It does not simply shift the entire waveform by a constant amount (that is a clamper). The teacher demonstrates positive and negative series/shunt, DC-biased and other waveform cases.



Illustrative ideal-diode waveforms

Classification	How to identify / solve
Series clipper	Diode in the series signal path; when OFF, no current flows in output resistor → v_o often 0 V.
Shunt clipper	Diode across the output/load; when ON, it provides a low-impedance path and clamps output toward reference level.
Positive clipper	Removes or limits the positive portion; whether named “positive” should be decided by clipped part.
Negative clipper	Removes or limits the negative portion.
Biased clipper	Added DC battery shifts the transition/clipping level from 0 to a threshold related to E (and 0.7 V if silicon).
Combinational clipper	Two appropriately oriented branches limit both positive and negative sides; check each diode independently.

The waveform-solving recipe (always draw these)

- Copy the exact circuit and output polarity from note.
- Mark the candidate diode states for v_i above and below the threshold (including any DC bias).
- Solve v_o using KVL with diode ON; solve again with it OFF.
- Find the switching value where a diode just changes state (e.g., $v_i = E$ for a relevant ideal biased circuit).

• Mark every transition point on the input and output axes; connect the correct pieces of the waveform.

Example 1 — ideal positive series clipper

If the diode orientation blocks positive input but conducts on negative input, then $v_o = 0$ when $v_i \geq 0$, and $v_o = v_i$ when $v_i < 0$. Reverse the diode and the retained half-cycle reverses. This matches the "positive clipped / negative clipped" waveform style in note pp. 8–10.

Example 2 — biased positive upper limiter

A shunt branch that turns ON and holds the output near +E gives (ideal diode) $v_o = \min(v_i, E)$, if the accompanying series resistor and polarities match this upper-limiter configuration. For a silicon diode in the corresponding direction, the upper clamp may be $\approx E+0.7$ V. Do not use the + sign without checking diode orientation and battery polarity.

Example 3 — two-sided ideal clipper

An upper limit of +3 V and a lower limit of −2 V produce: $v_o = -2$ V if $v_i < -2$ V; $v_o = v_i$ for $-2 \leq v_i \leq +3$ V; $v_o = +3$ V if $v_i > +3$ V. The output is a sine with both peaks flattened.

Matching book: §2.8 pp. 78–84; Examples 2.18–2.21 (pp. 80–83), especially Fig. 2.88, the clipper summary.

> Exam note: Your hand-drawn clippers often use IDEAL diodes and label "D ON: $v_o = 0$" or "D OFF: $v_o = v_i$." In the exam, follow the actual output node and element placement: ON does not universally mean zero output, especially when a battery is present.

9 | Clampers — output shifts without loss of swing (teacher pp. 15–16)

A clamper normally contains a capacitor in series with the input and a diode + resistor at the output. The capacitor charges while the diode conducts, then approximately holds charge as the diode turns OFF. The input waveform is shifted UP or DOWN by a DC amount.



Illustrative ideal-diode waveforms

> **Ideal clamper: $V_o = V_i - V_C$ (using V_C measured input-side plate minus output-side plate); peak-to-peak swing stays the same.**

In the first note-p. 15 square-wave circuit, a ±V input is clamped near 0 V during the +V portion and the other level becomes −2V. The second circuit reverses the diode; its output swings from 0 V to +2V. Your note p. 16 adds biased clampers, which shift the reference level again.

5-step procedure for the teacher's examples

• Mark one input half-cycle (typically the one that makes the diode ON).

• Replace the ON diode by its drop; solve capacitor voltage including polarity.

• Move to the other half-cycle; replace the diode by an open circuit.

• Retain approximately the same capacitor voltage and apply KVL for output.

• Check that V_out(max) − V_out(min) equals V_in(max) − V_in(min).

Worked example A — ideal square-wave clamper

Input alternates +5 V and −5 V (10 V peak-to-peak). If its positive level is clamped to 0 V, the other level is −10 V. Thus output levels are 0 V, −10 V; the 10 V peak-to-peak span is unchanged. If the negative level is instead clamped to 0 V, levels are 0 V, +10 V.

Worked example B — biased clamp with output maximum +3 V

For an ideal ±5-V input, fixing the output maximum at +3 V yields a −2-V shift, so output levels are +3 V and −7 V. Do not confuse clipping at +3 with clamping at +3; clipping would flatten a section rather than shift the full square wave.

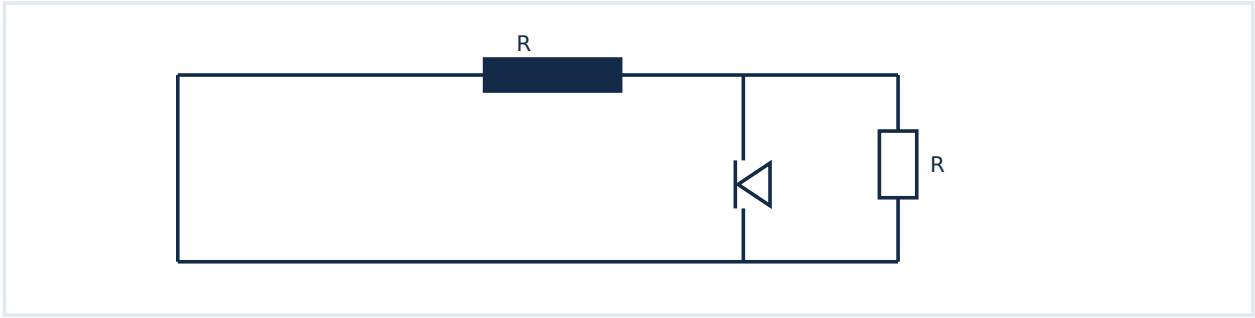
Why time constant matters (supporting textbook explanation)

Capacitor voltage should not drop much during diode-OFF period: choose RC ≫ signal half-period. The textbook checks approximately 5RC against the short duration of each interval. This is not a separate numerical topic in the class note, but explains why your drawn waveforms stay flat.

Matching book: §2.9 pp. 85–88; Examples 2.22–2.23; Figure 2.100 summarizes clampers.

10 | Zener diode and basic shunt regulator (teacher pp. 20–21)

Ordinary diode: forward conduction after ≈0.7 V (silicon). Zener diode: in reverse bias, after reaching its breakdown voltage V_Z, it conducts and holds approximately a constant voltage across itself and the parallel load—as long as there is enough Zener current and it stays within its rating.



Case I — V_i and R_L fixed; ALWAYS do the OFF-test first

> **Pretend Zener removed: V_test = V_i × R_L / (R + R_L).**

If V_test < V_Z: Zener OFF; ordinary resistor-divider equations apply. If V_test ≥ V_Z: test Zener ON using the equations below. At exact equality the Zener is at the ideal regulation boundary (I_Z ≈ 0); a practical device needs some minimum knee current.

> **Zener ON: V_L = V_Z; I_R = (V_i − V_Z)/R; I_L = V_Z/R_L; I_Z = I_R − I_L; P_Z = V_Z × I_Z.**

Worked example A — class note p. 21; book Example 2.26(a)

V_i = 16 V, R = 1 kΩ, R_L = 1.2 kΩ, V_Z = 10 V. Off-test V_test = (16×1.2)/(1+1.2) = 8.73 V, less than V_Z. Therefore OFF. V_L = 8.73 V; V_R = 16−8.73 = 7.27 V; I_R = I_L = 7.27 mA; I_Z = 0; P_Z = 0.

Worked example B — class note p. 21; book Example 2.26(b)

V_i = 16 V, R = 1 kΩ, R_L = 3 kΩ, V_Z = 10 V. Off-test V_test = 16×3/4 = 12 V; Zener ON. Then V_L=10 V; I_R = (16−10)/1k=6 mA; I_L=10/3k=3.33 mA; I_Z=2.67 mA; P_Z = 10×2.67mA=26.7 mW.

> Exam note: The OFF-test's 12 V in example B is hypothetical, not the FINAL output: once Zener turns ON, V_L falls to its regulated 10 V. Always recalculate currents after deciding the state.

11 | Zener advanced maths — the two exam variants (teacher p. 22)

Case II — V_i fixed; R_L variable (textbook Example 2.27)

As R_L decreases, more load current is needed. Regulation ends when Zener current reaches zero (ideal boundary). As R_L increases, load current falls and more current flows through Zener.

R_L(min) = R V_Z/(V_i − V_Z); I_R = (V_i−V_Z)/R; I_Z(max) = P_Z(max)/V_Z.

I_L(min) = I_R − I_Z(max); if I_L(min) > 0 then R_L(max) = V_Z/I_L(min).

If I_L(min) ≤ 0, the maximum Zener power is not reached even with no load; there is no finite R_L(max) imposed by that power limit. If a specific Zener knee current is provided, include it in the lower-load-bound calculation.

Worked example — book Example 2.27 (numbers unchanged)

V_i=50 V, R=1 kΩ, V_Z=10 V, allowed I_Z(max)=32 mA. R_L(min)=1k×10/(50−10)=250 Ω. I_R=40 mA. I_L(min)=40−32=8 mA. R_L(max)=10/0.008=1.25 kΩ. Permitted load range: 250 Ω to 1.25 kΩ, with ideal boundary at 250 Ω. P_Z(max)=10×32mA=320 mW.

Case III — R_L fixed; V_i variable (textbook Example 2.28)

Minimum supply allows the unloaded Zener test voltage to just reach V_Z. Maximum supply is determined by the largest allowed Zener current.

V_i(min) = V_Z × (R_L + R)/R_L.

I_L = V_Z/R_L; I_R(max) = I_Z(max)+I_L; V_i(max) = V_Z + I_R(max) × R.

Worked example — book Example 2.28

V_Z=20 V, R=220 Ω, R_L=1.2 kΩ, I_Z(max)=60 mA. V_i(min)=20×(1200+220)/1200=23.67 V. I_L=20/1200=16.67 mA. I_R(max)=60+16.67=76.67 mA. V_i(max)=20+(0.07667×220)=36.87 V. Ideal regulating range: 23.67–36.87 V.

> Exam note: The three regulator methods in the handwritten note exactly match textbook Example 2.26 (fixed-fixed), 2.27 (fixed V_i / varying load), and 2.28 (fixed load / varying input). These deserve repeated revision.

12 | Formula summary — memorize these

Area	What to write without thinking
ON Si	$V_D \approx 0.7$ V; current from KVL / Ohm
Single ON series diode	$I = (V_{supply,net} - 0.7) / R_{total}$
Load line	$I = (E - V_D)/R$; intercepts: E/R and E
Half-wave	$V_{DC} \approx 0.318 V_m$; PIV ≥ V_m; ripple f
Full bridge	$V_{DC} \approx 0.636 V_m$; PIV ≥ V_m; ripple 2f
Full CT	$V_{DC} \approx 0.636 V_m$ (per-half peak); PIV ≥ $2V_m$
Clamper	$V_{out,pp} = V_{in,pp}$; $V_o = V_i - V_C$ (defined polarity)
Zener OFF-test	$V_{test} = V_i R_L/(R+R_L)$
Zener ON	$V_L = V_Z$; $I_R = (V_i - V_Z)/R$; $I_L = V_Z/R_L$; $I_Z = I_R - I_L$
Zener variable load	$R_{Lmin} = RV_Z/(V_i - V_Z)$; $P_{Zmax} = V_Z I_{Zmax}$
Zener variable input	$V_{imin} = V_Z(1+R/R_L)$; $V_{imax} = V_Z + R(I_{Zmax} + V_Z/R_L)$

Approximation warning: the V_{DC} expressions are exact for ideal half/full sinusoidal outputs. Replacing V_m by $V_m - n(0.7\text{ V})$ provides a useful approximate silicon result when input peaks greatly exceed thresholds, but does not rigorously model the narrow turn-on region.

13 | Selected textbook exercises (only taught topics)

Use these problem numbers in the last ten pages of the textbook PDF. They are ordered in the same skill progression as the study guide, not by their order in the notebook. The problem figures in the book supply their circuit diagrams.

Skill	Book exercises (printed pages)	Difficulty / intention
Load lines	Problems 1–4 (pp. 119–120)	Q-point, ideal vs Si, target-current R
Series	Problems 5–9 (pp. 121–122)	ON/OFF, output voltages
Parallel & branching	Problems 10–13 (pp. 122–123)	KCL, multibranch currents
AND / OR gate	Problems 14–21 (p. 123)	Output levels and gate orientation
Half-wave rectifier	Problems 22–26 (pp. 123–124)	Sketch voltages / currents, DC
Bridge / CT full-wave	Problems 28–29 (pp. 124–125)	DC level, PIV, conduction paths
Clipper	Problems 32–36 (pp. 125–126)	Sketch biased and un-biased outputs
Clamper	Problems 37–39 (pp. 126–127)	Sketch output, RC meaning
Zener	Problems 42–44 (p. 128)	Variable R_L, variable V_i, regulation

> Exam note: Do not do every textbook exercise blindly. Start with the first one in each row, then the starred/most difficult one. Problems 30–31 and 40–41 are additional applications of rectification or clamper design: only attempt after completing your teacher’s direct circuit types.

14 | Your original practice set (attempt before the key)

Instructions: Unless specified, use ideal diodes for waveform questions and silicon $V_D = 0.7$ V for DC circuit questions. Sketch labelled axes and state when each diode conducts.

Q1 — load line $E = 12$ V and $R = 600\ \Omega$. Find both load-line intercepts. If ON Si, estimate Q-point.

Q2 — series A 9-V supply feeds a forward Si diode and a 1.5-kΩ resistor. Calculate V_R and I.

Q3 — reverse diode A 7-V source feeds a reverse-biased diode in series with 1-kΩ resistor. Find current and resistor voltage; give magnitude of diode voltage.

Q4 — two diodes A 15-V supply drives two forward silicon diodes in series with $R = 2.2$ kΩ. Find current.

Q5 — parallel sharing A 12-V source feeds 1 kΩ and two identical forward Si diodes in parallel. Find source current and assumed current in each diode.

Q6 — LED resistor A LED drops 2 V and needs 15 mA from a 9-V source. Find series R.

Q7 — gate logic List AND and OR outputs for input pairs 00, 01, 10, 11.

Q8 — gate analog A diode OR gate has an output pull-down and +5 V at A, 0 V at B. Approximate the silicon output voltage.

Q9 — half-wave Ideal half-wave rectifier has $V_m = 15$ V. Find V_{DC} and diode PIV.

Q10 — bridge Ideal bridge has $V_m = 24$ V. Find V_{DC}, diode PIV and output ripple frequency if input is 50 Hz.

Q11 — CT An ideal CT secondary gives 10 V rms from each end to the center tap. Find V_m per half, V_{DC} and PIV per diode.

Q12 — transformer An ideal transformer has $N_p:N_s = 5:1$ and primary 230 V rms. Find entire secondary rms voltage.

Q13 — positive clipper An ideal positive series clipper blocks positive input. For v_i values −4, 0, +3 V, write the three v_o values.

Q14 — upper limiter Ideal output upper-limited at +3 V. Find v_o for $v_i = -6, +1, +5$ V.

Q15 — dual clipper Ideal limits −2 V and +4 V. Find output for inputs −6, +1, +7 V.

Q16 — clamper Ideal ±4 V square input: positive level clamped to 0 V. Find both output levels and V_{pp}.

Q17 — biased clamper Ideal ±6 V square wave has maximum output +2 V. Find output minimum.

Q18 — Zener OFF $V_i=15$ V, $R=1$ kΩ, $R_L=1$ kΩ, $V_Z=10$ V. Determine ON/OFF and V_L.

Q19 — Zener ON $V_i=18$ V, $R=1$ kΩ, $R_L=2$ kΩ, $V_Z=9$ V. Find V_L, I_R, I_L, I_Z, P_Z.

Q20 — Zener variable load $V_i=30$ V, $R=500\ \Omega$, $V_Z=10$ V, $I_{Z(max)}=30$ mA. Find $R_{L(min)}$, $R_{L(max)}$ and $P_{Z(max)}$.

Q21 — Zener variable input $R=300\ \Omega$, $R_L=1.5$ kΩ, $V_Z=12$ V, $I_{Z(max)}=40$ mA. Find the ideal regulating V_i range.

Q22 — diode model Explain in two sentences why a reverse-biased diode may have large voltage yet zero current.

15 | Practice answer key — show your workings in the test

Q1 I-axis intercept = 12/600 = 20 mA; V-axis intercept = 12 V. Si estimate: $V_D \approx 0.7$ V, $I \approx 11.3/600 = 18.83$ mA.

Q2 $V_R = 9 - 0.7 = 8.3$ V; $I = 8.3/1500 = 5.53$ mA.

Q3 $I = 0$, $V_R = 0$; reverse diode $|V_D| = 7$ V.

Q4 $I = [15-(2\times0.7)]/2200 = 6.18$ mA.

Q5 Total $I = (12-0.7)/1000 = 11.3$ mA; each identical diode ≈ 5.65 mA.

Q6 $R = (9-2)/0.015 = 466.7\ \Omega$; choose nominal ≈470 Ω if a standard resistor is required.

Q7 AND: 0,0,0,1; OR: 0,1,1,1.

Q8 Output ≈ 5−0.7 = 4.3 V (logic HIGH).

Q9 $V_{DC} = 0.318\times15 = 4.77$ V; diode PIV ≥15 V.

Q10 $V_{DC} = 0.636\times24 = 15.26$ V; PIV ≥24 V; ripple = 100 Hz.

Q11 $V_m = 10\sqrt{2} = 14.14$ V per half; $V_{DC} = 0.636 \times 14.14 = 9.00$ V; PIV $\geq 2 \times 14.14 = 28.28$ V.

Q12 Secondary = 230/5 = 46 V rms.

Q13 $v_o = -4, 0, 0$ V.

Q14 $v_o = -6, +1, +3$ V.

Q15 $v_o = -2, +1, +4$ V.

Q16 Output = 0 and −8 V; $V_{pp} = 8$ V.

Q17 Input $V_{pp} = 12$ V; minimum = +2−12 = −10 V.

Q18 $V_{test} = 15 \times 1/(1+1) = 7.5$ V <10 V; Zener OFF; $V_L = 7.5$ V.

Q19 $V_{test} = 18 \times 2/3 = 12 > 9$, ON. $V_L = 9$ V; $I_R = (18-9)/1k = 9$ mA; $I_L = 9/2k = 4.5$ mA; $I_Z = 4.5$ mA; $P_Z = 40.5$ mW.

Q20 $R_{Lmin} = 500 \times 10/(30-10) = 250\ \Omega$; $I_R = 20/500 = 40$ mA; $I_{Lmin} = 40-30 = 10$mA; $R_{Lmax} = 10/0.01 = 1\ k\Omega$; $P_{Zmax} = 300$mW.

Q21 $V_{imin} = 12(1+300/1500) = 14.4$ V. $I_L = 12/1500 = 8$ mA; $I_{Rmax} = 48$mA; $V_{imax} = 12 + 0.048 \times 300 = 26.4$V.

Q22 An OFF diode is modeled as an open circuit; therefore I=0. KVL allows the supply voltage to appear across the open device.

16 | Seven-day exam study plan

Recommended if you can study roughly 2–3 focused hours per day. Shorter sessions are fine—keep the same order and divide a day into two. Spend 40% reading/redrawing notes, 15% matching book examples, 45% active problem-solving.

Day	Main study	You must produce
1	Diode ON/OFF, KVL/KCL, load line; note p. 1–3; book §2.2–2.3.	1 clean load-line graph; solve Q1–4 and book 1–6.
2	Parallel / series-parallel; gates; note pp. 4–7; book §2.4–2.5.	Current branch sketches, AND+OR circuits/truth tables; Q5–8.
3	Half-wave, bridge, CT transformer; notes 17–19; book §2.6–2.7.	Three circuits and six labelled waveforms; Q9–12.
4	Clippers; notes 8–14; book §2.8.	Draw 4 basic clipper outputs + 2 biased ones; Q13–15.
5	Clampers; notes 15–16; book §2.9.	Draw 4 output-level diagrams and check V_{pp}; Q16–17.
6	Zener; notes 20–22; book §2.11.	Rework book Examples 2.26–2.28 without solutions; Q18–21.
7	Mixed mock test, mistakes, final formula recall.	Finish mock below under a time limit; rewrite every wrong answer.

Three-day emergency plan

Day A: diode methods + networks + gates (notes 1–7). Day B: rectifiers + clippers + clampers (notes 8–19). Day C: Zener 3 cases (20–22) + mock test + formula review. Spend more time on Zener and waveforms than reading narrative paragraphs.

Simple checklist for every topic

☐ I can redraw the exact circuit from memory.

☐ I know which diodes are ON and OFF in each input condition.

☐ I can derive the voltage/current maths with units.

☐ I can draw input and output axes with the correct voltage levels.

☐ I can explain the result in 1–2 sentences.

☐ I can solve an unseen, similar question without notes.

17 | Mock class test (60 minutes, 40 marks)

Q1. Short conceptual answers (8 marks) — Define Q-point (2). State ideal vs practical Si model (2). Differentiate clipper vs clamper (2). Explain diode PIV (2).

Q2. Diode network calculation (7 marks) — A 12-V battery, a conducting Si diode and a 2.2-kΩ resistor are in series: find diode voltage, resistor voltage, current (4). Two identical such diodes are connected in parallel behind one 2.2-kΩ resistor: find each diode current under equal-sharing assumption (3).

Q3. Diode logic (5 marks) — Draw diode AND and diode OR arrangements with their output pull resistors, and supply their truth tables.

Q4. Waveform analysis (8 marks) — Show the positive and negative portions of the output of (a) an ideal half-wave rectifier and (b) a bridge rectifier (4). For $V_m=20$ V, find V_{DC} of both and the individual PIV requirements (4).

Q5. Waveform shaping (6 marks) — An ideal limiter clips at +4 V, input sine peaks at ±8 V: sketch output (3). An ideal clamper receives a ±5-V square wave and clamps its negative level at −2 V: find the other output level and sketch output (3).

Q6. Zener maths (6 marks) — $V_i=20$ V, $R=1$ kΩ, $R_L=2$ kΩ, $V_Z=9$ V. Test ON/OFF (2), then find V_L, I_R, I_L, I_Z (4).

Mock marking key (self-check only after attempting)

Q2: $V_D=0.7$ V; $V_R=11.3$ V; $I=5.14$ mA; parallel identical diode ≈2.57 mA each. Q4: half $V_{DC}=6.36$ V/PIV ≥20 V; bridge $V_{DC}=12.72$ V/PIV ≥20 V. Q5: upper clamp max +4 V, negative reaches −8 V; square input span 10 V so other level is +8 V. Q6: $V_{test}=20\times2/3=13.33$ V >9 so ON; $V_L=9$ V; $I_R=11$ mA; $I_L=4.5$ mA; $I_Z=6.5$ mA. Q1/Q3 marked by definitions/drawings/truth tables.

> Exam note: A 40-mark, 60-minute mock is a practice format created for this guide; your teacher has not supplied a test duration or mark distribution in these PDFs.

18 | Ten recurring exam mistakes to eliminate

01. Using 0.7 V for a diode that is actually OFF.
02. Forgetting OFF means zero branch current, not zero diode voltage.
03. Getting the signs of negative DC sources wrong in a KVL loop.
04. Dividing total current equally across unequal parallel diodes without a stated assumption.
05. Treating logic HIGH as always exactly +5 V after a silicon OR diode.
06. Switching positive/negative clipper names without checking what half is removed.
07. Treating a clamper as a waveform cutter; forgetting V_{pp} is preserved.
08. Confusing rms and peak: $V_m = \sqrt{2}\, V_{rms}$ for sinusoidal AC.
09. Using $2V_m$ PIV for a bridge or V_m PIV for CT without defining V_m.
10. Using the Zener OFF-test output as the final ON-regulated output.

End-of-guide final target: Be able to explain the circuit state, write the equation, complete the numerical substitution and sketch the output waveform for every diagram on the 22 notebook pages.